

Exploratory Data
Analysis and
Statistical
Investigation of the
Stack Overflow
Developer Survey
Dataset

1. Dataset Summary

1.1 Introduction

The Stack Overflow Developer Survey dataset contains responses collected from software developers worldwide. The survey gathers information about education, employment status, programming languages, technologies used, salary, experience, and developer preferences. This dataset is widely used for understanding trends in the software development industry.

1.2 Dataset Size

Attribute	Value
Number of Rows	49191
Number of Columns	172
File Format	CSV
Missing Values	4447043
Duplicate Records	0

1.3 Key Features

Feature	Description
Age	Age group of respondent
Country	Country of residence
YearsCode	Professional coding experience
Employment	Employment status
LanguageHaveWorkedWith	Programming languages used
ConvertedCompYearly	Annual salary in USD

1.4 Dataset Snapshot

df.head()

	index	ResponseId	MainBranch	Age	...	AIOpen	ConvertedCompYearly	JobSat	Random
0	0.0	1.0	I am a developer by profession	25-34 years old	...	Troubleshooting, profiling, debugging	61256.0	10.0	22.0
1	1.0	2.0	I am a developer by profession	25-34 years old	...	All skills. AI is a flop.	104413.0	9.0	12.0
2	2.0	3.0	I am a developer by profession	35-44 years old	...	Understand how things actually work, problem s...	53061.0	8.0	22.0
3	3.0	4.0	I am a developer by profession	35-44 years old	...	NaN	36197.0	6.0	9.0
4	4.0	5.0	I am a developer by profession	35-44 years old	...	critical thinking, the skill to define the tas...	60000.0	7.0	43.0

5 rows × 174 columns

2. Data Exploration Plan

2.1 Purpose of Exploration

The purpose of exploratory data analysis is to understand the structure, quality, and characteristics of the Stack Overflow Developer Survey dataset. The analysis aims to identify patterns, trends, relationships, and anomalies within the data before applying statistical methods or predictive modeling techniques.

2.2 Planned Exploratory Techniques

Objective	Technique
Understand distributions	Histograms
Compare categories	Bar Charts
Compare salary groups	Box Plots
Study relationships	Scatter Plots
Measure associations	Correlation Heatmap
Detect outliers	Box Plots & IQR

2.3 Data Quality Assessment Plan

Before conducting analysis, the dataset will be examined for:

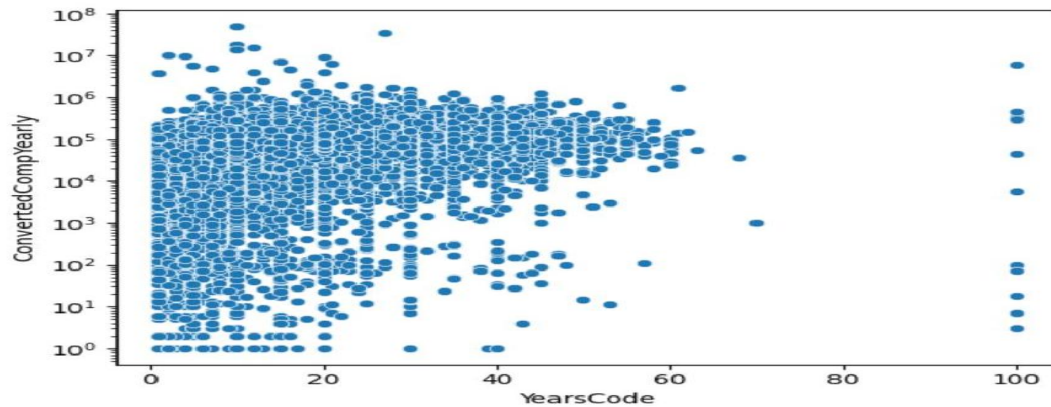
- Missing values
- Duplicate records
- Inconsistent entries
- Outliers
- Incorrect data types

2.4 Expected Outcomes

Through exploratory analysis, the study aims to uncover trends in developer demographics, technology preferences, and compensation patterns. The findings will be used to formulate hypotheses and perform statistical testing in later sections of the report.

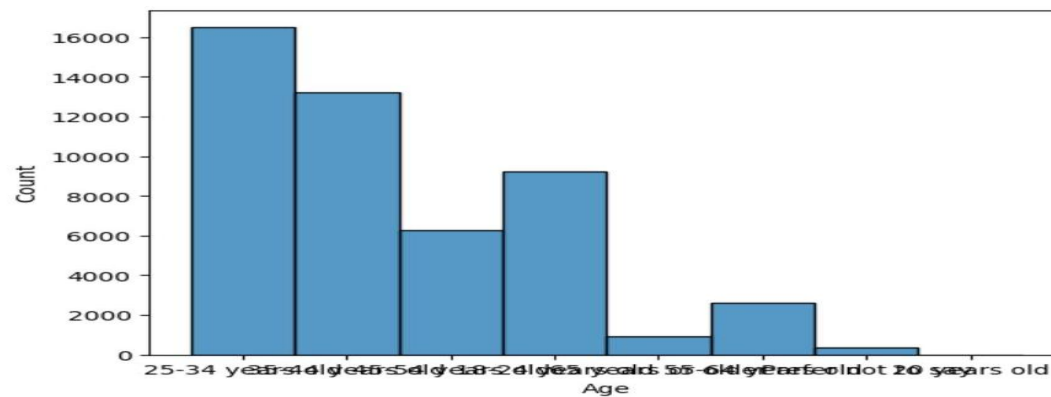
3. Exploratory Data Analysis

```
sns.scatterplot(x='YearsCode', y='ConvertedCompYearly', data=df)
plt.yscale('log')
plt.show()
```

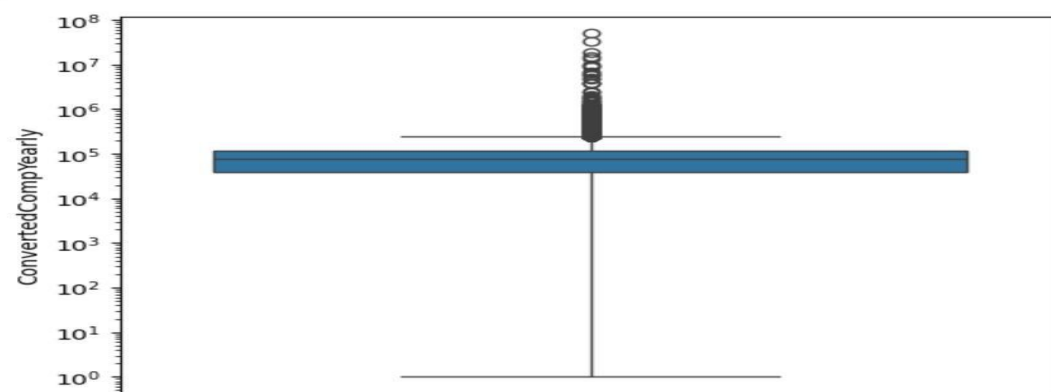


```
sns.histplot(df['Age'])
```

<Axes: xlabel='Age', ylabel='Count'>



```
sns.boxplot(df['ConvertedCompYearly'])
plt.yscale('log')
plt.show()
```



Key Findings

- The age distribution is concentrated within a few dominant age groups.
- Several salary outliers are present in the dataset.
- Professional experience shows a positive relationship with annual compensation.

4. Data Cleaning and Feature Engineering

4.1 Missing Value Treatment

Missing values were identified and handled using appropriate techniques. Rows containing critical missing information were removed, while non-critical missing values were retained or imputed where necessary.

4.2 Duplicate Data Check

The dataset was examined for duplicate records. However no duplicate values were found.

4.3 Data Type Conversion

Relevant columns were converted to appropriate data types to facilitate analysis and visualization.

Example:

YearsCode → numeric

ConvertedCompYearly → numeric

4.4 Outlier Treatment

Salary data contained several extreme values. Outliers were identified using boxplot analysis.

4.5 Feature Engineering

A logarithmic transformation of annual compensation was created to reduce skewness and improve interpretability of salary distributions.

$\text{LogSalary} = \log(\text{ConvertedCompYearly})$

4.6 Summary

Data cleaning involved handling missing values, checking for duplicates, converting data types, and examining outliers. Feature engineering included creating derived variables to support exploratory and statistical analysis.

5. Key Findings and Insights

- Most respondents belong to a few dominant age groups.
- Salary distribution is heavily right-skewed with several high-income outliers.
- Professional experience is positively associated with annual compensation.
- Compensation varies considerably among respondents with similar experience levels.
- Additional factors beyond experience likely influence salary outcomes.

6. Hypothesis Formulation

H0: There is no relationship between professional experience and annual compensation.

H1: Professional experience has a significant positive relationship with annual compensation.

Conclusion

This report analyzed the Stack Overflow Developer Survey dataset using exploratory data analysis and data cleaning techniques. The analysis revealed that annual compensation is positively associated with professional experience and that salary distributions are highly right-skewed with the presence of outliers. The findings provide valuable insights into developer demographics and compensation patterns within the dataset.

References

Help from online resources and ai has been taken in taking guidance and adding general descriptions